

Zonal based tracking system

Frontend

- Link : <https://tracker-backend-ten.vercel.app/>
- Tech stack used:
 - HTML5
 - CSS3
 - Bootstrap
 - JavaScript ES6
- Deployed using vercel

Backend

- Link: <https://tracker-backend-1djh.onrender.com/devices>
- Tach stack used :
 - NodeJS
 - Express
- Deployed using Render

Receiver antenna

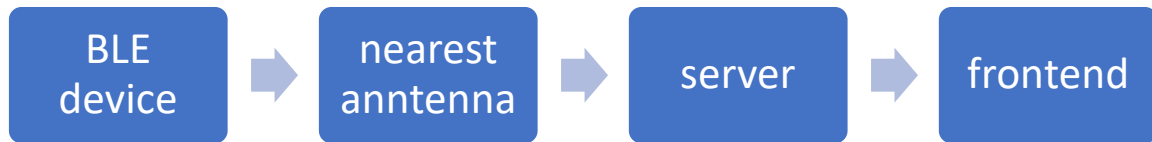
- Device used Laptops(2)
- Tach stack used :
 - Python
 - Bleak, requests

BLE Advertiser's

- Device used phones
- BLE Advertiser APP

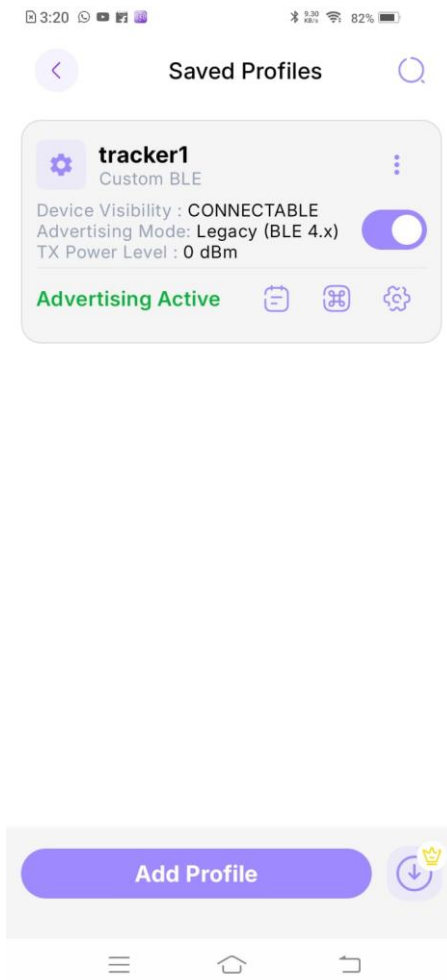
Git hub link:

https://github.com/akshayAK25/tracker_backend



BLE device (BLE Advertiser)

Advertising it self continuously



Python code (receiver antenna)

Detecting multiple BLE devices tracking there RSSI values and sending the data to server along with its receiver id (Laptop1).

```
PS C:\Users\AKSHAY\OneDrive\Desktop\tracker project> python scanner.py

--- TRACKERS FOUND ---
{'receiver': 'Laptop1', 'device': 'tracker1', 'rssi': -57, 'location': 'Hall'}
Sent: 200
{'receiver': 'Laptop1', 'device': 'tracker2', 'rssi': -71, 'location': 'Hall'}
Sent: 200

--- TRACKERS FOUND ---
{'receiver': 'Laptop1', 'device': 'tracker1', 'rssi': -63, 'location': 'Hall'}
Sent: 200
{'receiver': 'Laptop1', 'device': 'tracker2', 'rssi': -45, 'location': 'Hall'}
Sent: 200

--- TRACKERS FOUND ---
{'receiver': 'Laptop1', 'device': 'tracker1', 'rssi': -47, 'location': 'Hall'}
Sent: 200
{'receiver': 'Laptop1', 'device': 'tracker2', 'rssi': -45, 'location': 'Hall'}
Sent: 200
[]
```

Server (NodeJS)

Receives data from multiple antennas (python code) about multiple BLE Advertisers, there RSSI values and Receivers id.

```
tracker-backend-1djh.onrender.com/devices

Pretty-print ✓

{
  "tracker1": {
    "currentLocation": "Bedroom",
    "bestRSSI": -40,
    "nearestReceiver": "Laptop2",
    "status": "Very Near",
    "receivers": {
      "Laptop2": {
        "location": "Bedroom",
        "rssi": -40,
        "timestamp": "2026-05-28T08:44:59.702Z"
      },
      "Laptop1": {
        "location": "Hall",
        "rssi": -47,
        "timestamp": "2026-05-28T08:46:28.233Z"
      }
    }
  },
  "tracker2": {
    "currentLocation": "Bedroom",
    "bestRSSI": -32,
    "nearestReceiver": "Laptop2",
    "status": "Very Near",
    "receivers": {
      "Laptop2": {
        "location": "Bedroom",
        "rssi": -32,
        "timestamp": "2026-05-28T09:08:32.477Z"
      },
      "Laptop1": {
        "location": "Hall",
        "rssi": -49,
        "timestamp": "2026-05-28T09:31:08.158Z"
      }
    }
  }
}
```

Frontend Application

Receives data from the server and display it accordingly.

